Trace heights on quadratic twists with $j = 1728$

13 November 2026

Abstract

For a quadratic number field K and $d \in K^\times$ with $d \neq \sigma(d)$ and $d \neq -\sigma(d)$, the rational trace of the abscissa on $y^2 = x^3 - d^2x$ controls the canonical height of every K-rational point with finite abscissa. The additive constant is ineffective. The proof excludes all positive-dimensional abelian translates in the relevant finite fibers and checks the projective embedding required by von Buhren's family height theorem. No rank-one hypothesis or square-abscissa condition is needed for this height result. For the curve used in the Diophantine construction, the conclusion therefore covers half-integral indices and torsion translates. We also supply a reciprocal-coordinate description of an ordered pair of points and its difference, including the identity and tangent cases.

1 The height theorem

Heights are absolute logarithmic Weil heights. For $a/b \in \mathbb{Q}$ reduced, $b > 0$, write $h(a/b) = \log \max(|a|, b)$. Canonical heights on elliptic curves use the standard symmetric divisor at the identity; changing its normalization only changes the positive constants below.

Theorem 1.1. *Let $K/\mathbb{Q}$ be quadratic with nontrivial automorphism σ. Let $d \in K^\times$ satisfy $d \neq \sigma(d)$ and $d \neq -\sigma(d)$, and set $E_d : y^2 = x^3 - d^2x$. There are constants $A > 0$ and $J \geq 0$, depending only on K, d, such that every $Q \in E_d(K) \setminus \{O\}$ satisfies*

$$\widehat{h}(Q) \leq Ah(\mathrm{Tr}_{K/\mathbb{Q}}\, x(Q)) + J.$$

The theorem is unconditional. The argument does not give an effective numerical value of J.

Lemma 1.2. *On E_0^2, where $E_0 : Y^2 = X^3 - X$, let $F(Q_1, Q_2) = ax(Q_1) + bx(Q_2)$ with $ab \neq 0$ and $a \neq \pm b$. No translate of a positive-dimensional abelian subvariety which meets $(E_0 \setminus \{O\})^2$ has constant restriction of F.*

Proof. The whole surface does not have constant F. A proper positive-dimensional abelian subvariety is an elliptic curve. If one projection of its translate is constant, the other projection is nonconstant and F is not constant. Otherwise, after an isogeny from E_0 to this elliptic curve, the two projections have the form $\alpha z + q_1$ and $\beta z + q_2$, with nonzero endomorphisms α, β of E_0.

If F were constant, its two summands would have identical pole supports. Their poles are all double, since the isogenies are etale in characteristic zero. Choose a common pole and translate the parameter so it is zero. The projections become $\alpha z, \beta z$; the equality of pole supports gives $\ker \alpha = \ker \beta$. Factoring through their common quotient gives $\beta = u\alpha$ for an origin-preserving automorphism u of E_0. On this short Weierstrass model such an automorphism sends x to x or $-x$. For example, its action on the linear system $|2O|$ is affine in x; preservation of the three finite branch points $-1, 0, 1$ forces its translation to be zero and its slope to be 1 or -1. Consequently the restriction is $(a + b)x(\alpha z)$ or $(a - b)x(\alpha z)$, a nonconstant function under the hypotheses. □

Proof of Theorem 1.1. Choose square roots of d and $d' = \sigma(d)$ in a fixed number-field extension. The pointed isomorphisms to E_0 send (x, y) to $(x/d, y/d^{3/2})$ and $(x/d', y/(d')^{3/2})$, respectively. The image of a point and its conjugate is therefore a point of E_0^2 at which

$$F = dx_1 + d'x_2 = \operatorname{Tr}_{K/\mathbb{Q}} x(Q).$$

Take the integral reduced projective closure $\mathcal{V}$ of the graph of F in $E_0^2 \times \mathbf{P}^1$. Its dimension is two. The set of images of $E_d(K)$ is a finitely generated subgroup Γ of $E_0^2(\overline{\mathbb{Q}})$, by Mordell–Weil and the fact that the two pointed maps commute with the group operations.

We apply the published Theorem 1.1 of von Buhren [1]. It bounds canonical height linearly in the parameter height, with a fixed additive constant for each finite-rank subgroup, away from the union of positive-dimensional abelian translates contained in the fiber. A point of the graph above an actual finite value of F lies outside that union: a translate through it would meet the domain of F in a dense open subset and make F constant there, contrary to Lemma 1.2. This also treats possible components of the graph over the indeterminacy locus; none changes the argument through our point in the domain.

Here are the remaining embedding and subgroup checks. Embed E_0 as its smooth plane cubic by $\mathcal{O}(3O)$, and embed its square by Segre in $\mathbf{P}^8$. For $n \geq 1$ the sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^2}(n-3) \longrightarrow \mathcal{O}_{\mathbf{P}^2}(n) \longrightarrow \mathcal{O}_{E_0}(3nO) \longrightarrow 0$$

and $H^1(\mathbf{P}^2, \mathcal{O}(n-3)) = 0$ prove surjectivity of the restriction of sections. Tensoring the two surjections and using Segre proves condition (1.1) in von Buhren's statement for every $n \geq 1$. The line bundle on the surface is ample and symmetric. On the base use $\mathcal{O}_{\mathbf{P}^1}(1)$ with its ordinary height. Every point of Γ lies in the source set $C(\Gamma, \epsilon)$ by taking its error point to be zero. Thus no uniformity in a varying group or field is being invoked.

The canonical height for the chosen product embedding is a fixed positive multiple of the sum of the two elliptic canonical heights. Pointed isomorphism invariance and Galois invariance identify these with the canonical height of Q. The source theorem now gives the displayed inequality. Its additive constant depends on this fixed Γ and is not made effective by the source. Increase constants if necessary to make $A > 0$ and $J \geq 0$. □

Corollary 1.3. *Let $K = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$, and $P = 2R_0$, where $R_0 = (1 + \sqrt{2}, 2\sqrt{2})$. Retain the arithmetic premise $2E_d(K) = \mathbb{Z}P$ with P non-torsion. Every $Q \in E_d(K)$ has a unique index $n \in \frac{1}{2}\mathbb{Z}$, defined by $Q = [2n]R_0 + T_0$ with $2T_0 = O$. There are fixed $A > 0, J \geq 0$ such that*

$$n^2 \leq A h(\operatorname{Tr} x(Q)) + J \qquad (Q \neq O).$$

This includes finite torsion points, with index zero.

Proof. If $2Q = mP$, then $2(Q - mR_0) = O$. Uniqueness follows from non-torsion of R_0. The canonical height of Q equals $n^2 \widehat{h}(P)$. Apply the theorem and divide by its positive constant $\widehat{h}(P)$; $d + d' = 6$ and $d - d' = -2\sqrt{2}$ are nonzero. □

2 Two reciprocal points and their difference

This section is algebraic and does not assume a group-rank statement. Work over any field of characteristic zero, with $D \neq 0$, on $E : y^2 = x^3 - Dx$. The rational functions

$$f = 1/x, \qquad h = -y/x^2$$

extend by $(f, h) = (0, 0)$ at O. They identify $E \setminus \{(0,0)\}$ with $h^2 = f - Df^3$. Indeed the affine map $(f, h) \mapsto (-Df, -Dh)$ gives translation by $(0, 0)$ on E.

Proposition 2.1. *The three equations*

$$h^2 = f - Df^3, \tag{1}$$
$$s(f-g) = h + k, \tag{2}$$
$$s(h-k) = 1 - D(f^2 + fg + g^2) \tag{3}$$

describe ordered pairs $U, V \in E \setminus \{(0,0)\}$ with $U \neq V$, and their unique secant or tangent parameter s. Their difference is the finite point

$$A = U - V = (c, y_A), \qquad c = s^2 + D(f+g), \quad y_A = Dh - s(Df + c).$$

Either U or V may be O. Finite torsion differences are permitted.

Proof. Put $r_1 = s(f-g) - h - k$, $r_2 = s(h-k) - 1 + D(f^2 + fg + g^2)$ and $r_3 = h^2 - f + Df^3$. Exact polynomial identities give

$$k^2 - g + Dg^3 = r_3 - (f-g)r_2 + (h-k)r_1,$$
$$y_A^2 - c^3 + Dc = D^2 r_3 - D(Df + c)(r_2 - sr_1).$$

Thus both reciprocal points and the asserted finite point are on the curve. If $f \neq g$, translate both points by $(0,0)$. The line joining the first translated point to the negative of the second has slope s; the chord law gives exactly (c, y_A). Translation by this two-torsion point does not alter their difference.

If $f = g$, the equations give $k = -h$. If also $h = 0$, the cubic gives $f = 0$ or $Df^2 = 1$, whereas (3) would give $1 - 3Df^2 = 0$, a contradiction in either case. Hence $h \neq 0$ and $2sh = 1 - 3Df^2$. This is the tangent slope at the first translated point, so the tangent law gives the same formula. In particular $U \neq V$ in every case. Conversely these secant and tangent calculations supply s for every allowed ordered pair. The identity case is the ordinary affine point $(f, h) = (0, 0)$ in this translated model and is already covered. □

3 Ratio equations and formal precision

If H is finite with $S = y(H)/x(H) \neq 0$, its formal parameter is $z(H) = -1/S$. The two equations

$$hL + fS = 0, \qquad (1 - Df^2)L + hS = 0$$

give $L = z(U)/z(H)$ whenever U is non-torsion or O; at O the second equation gives $L = 0$. At a nonidentity formal point $h \neq 0$, so the first equation gives the ratio and the second follows from the cubic. We impose both equations in the quartic construction and do not divide by h on other components.

For completeness, at a good unramified prime $p > 5$, D a unit, the invariant differential and formal logarithm in $z = -x/y$ are

$$\omega = \frac{dz}{\sqrt{1 + 4Dz^4}}, \qquad \log_E(z) = \sum_{j \geq 0} \frac{(-1)^j \binom{2j}{j} D^j}{4j+1} z^{4j+1}.$$

The first identity follows from $x = (1 + \sqrt{1 + 4Dz^4})/(2z^2)$. The binomial coefficients are integers. For $j \geq 1$, $v_p(4j+1) \leq 4j - 4$, so if $v_p(z) \geq 1$ all omitted terms have relative valuation at least $4v_p(z)$. Consequently $\log_E(z) = z(1 + O(p^{4v_p(z)}))$. At O both logarithm and parameter are zero; the multiplicative unit is assigned the value one. This supplies the order-four relative precision used by the new quartic without imposing a positive lower bound on a numerator index.

4 Verification and source scope

The exact code accompanying the 81-variable quartic checks the two polynomial identities for the pair symbolically. A separate standard-library program checks 702 ordered pairs over $\mathbb{Q}(\sqrt{2})$, including 24 tangent pairs, 52 pairs involving the identity, 78 nonzero torsion differences, and 352 half-integral index differences. It also checks 1300 regular ratio instances. These are finite examples, not a substitute for the universal proof or for Mordell–Weil.

The final published statement of von Buhren's Theorem 1.1 and its embedding condition were checked in text and on the image of printed page 308. The application above is proved here; the internal proof of the published theorem is not represented as independently re-audited. No *abc*, prime-pattern or pole-realization hypothesis enters Theorem 1.1; the rank premise enters only its index corollary.

References

[1] Jérôme von Buhren, *Borne de hauteur semi-effective pour le problème de Mordell–Lang dans une variété abélienne*, Journal de Théorie des Nombres de Bordeaux 29 (2017), 307–320, Theorem 1.1, p. 308. DOI: 10.5802/jtnb.981. https://www.numdam.org/article/JTNB_2017__29_1_307_0.pdf.